

we assume the model outputs only the confidence for the positive class.¹ Given a sample $\mathbf{x}_i$, we have access to $\hat{p}_i$ – the network’s predicted probability of $y_i = 1$, as well as $z_i \in \mathbb{R}$ – which is the network’s non-probabilistic output, or *logit*. The predicted probability $\hat{p}_i$ is derived from z_i using a sigmoid function σ ; i.e. $\hat{p}_i = \sigma(z_i)$. Our goal is to produce a calibrated probability $\hat{q}_i$ based on y_i , $\hat{p}_i$, and z_i .

Histogram binning (Zadrozny & Elkan, 2001) is a simple non-parametric calibration method. In a nutshell, all uncalibrated predictions $\hat{p}_i$ are divided into mutually exclusive bins $B_1, \dots, B_M$. Each bin is assigned a calibrated score θ_m ; i.e. if $\hat{p}_i$ is assigned to bin B_m , then $\hat{q}_i = \theta_m$. At test time, if prediction $\hat{p}_{te}$ falls into bin B_m , then the calibrated prediction $\hat{q}_{te}$ is θ_m . More precisely, for a suitably chosen M (usually small), we first define bin boundaries $0 = a_1 \leq a_2 \leq \dots \leq a_{M+1} = 1$, where the bin B_m is defined by the interval $(a_m, a_{m+1}]$. Typically the bin boundaries are either chosen to be equal length intervals or to equalize the number of samples in each bin. The predictions θ_i are chosen to minimize the bin-wise squared loss:

$$\min_{\theta_1, \dots, \theta_M} \sum_{m=1}^M \sum_{i=1}^n \mathbf{1}(a_m \leq \hat{p}_i < a_{m+1}) (\theta_m - y_i)^2, \quad (7)$$

where $\mathbf{1}$ is the indicator function. Given fixed bins boundaries, the solution to (7) results in θ_m that correspond to the average number of positive-class samples in bin B_m .

Isotonic regression (Zadrozny & Elkan, 2002), arguably the most common non-parametric calibration method, learns a piecewise constant function f to transform uncalibrated outputs; i.e. $\hat{q}_i = f(\hat{p}_i)$. Specifically, isotonic regression produces f to minimize the square loss $\sum_{i=1}^n (f(\hat{p}_i) - y_i)^2$. Because f is constrained to be piecewise constant, we can write the optimization problem as:

$$\begin{aligned} \min_{\substack{\theta_1, \dots, \theta_M \\ a_1, \dots, a_{M+1}}} & \sum_{m=1}^M \sum_{i=1}^n \mathbf{1}(a_m \leq \hat{p}_i < a_{m+1}) (\theta_m - y_i)^2 \\ \text{subject to} & \quad 0 = a_1 \leq a_2 \leq \dots \leq a_{M+1} = 1, \\ & \quad \theta_1 \leq \theta_2 \leq \dots \leq \theta_M. \end{aligned}$$

where M is the number of intervals; $a_1, \dots, a_{M+1}$ are the interval boundaries; and $\theta_1, \dots, \theta_M$ are the function values. Under this parameterization, isotonic regression is a strict generalization of histogram binning in which the bin boundaries and bin predictions are jointly optimized.

Bayesian Binning into Quantiles (BBQ) (Naeini et al., 2015) is an extension of histogram binning using Bayesian

¹ This is in contrast with the setting in Section 2, in which the model produces both a class prediction and confidence.

model averaging. Essentially, BBQ marginalizes out all possible *binning schemes* to produce $\hat{q}_i$. More formally, a binning scheme s is a pair $(M, \mathcal{I})$ where M is the number of bins, and $\mathcal{I}$ is a corresponding partitioning of $[0, 1]$ into disjoint intervals $(0 = a_1 \leq a_2 \leq \dots \leq a_{M+1} = 1)$. The parameters of a binning scheme are $\theta_1, \dots, \theta_M$. Under this framework, histogram binning and isotonic regression both produce a single binning scheme, whereas BBQ considers a space $\mathcal{S}$ of all possible binning schemes for the validation dataset D . BBQ performs Bayesian averaging of the probabilities produced by each scheme:²

$$\begin{aligned} \mathbb{P}(\hat{q}_{te} \mid \hat{p}_{te}, D) &= \sum_{s \in \mathcal{S}} \mathbb{P}(\hat{q}_{te}, S = s \mid \hat{p}_{te}, D) \\ &= \sum_{s \in \mathcal{S}} \mathbb{P}(\hat{q}_{te} \mid \hat{p}_{te}, S = s, D) \mathbb{P}(S = s \mid D). \end{aligned}$$

where $\mathbb{P}(\hat{q}_{te} \mid \hat{p}_{te}, S = s, D)$ is the calibrated probability using binning scheme s . Using a uniform prior, the weight $\mathbb{P}(S = s \mid D)$ can be derived using Bayes’ rule:

$$\mathbb{P}(S = s \mid D) = \frac{\mathbb{P}(D \mid S = s)}{\sum_{s' \in \mathcal{S}} \mathbb{P}(D \mid S = s')}.$$

The parameters $\theta_1, \dots, \theta_M$ can be viewed as parameters of M independent binomial distributions. Hence, by placing a Beta prior on $\theta_1, \dots, \theta_M$, we can obtain a closed form expression for the marginal likelihood $\mathbb{P}(D \mid S = s)$. This allows us to compute $\mathbb{P}(\hat{q}_{te} \mid \hat{p}_{te}, D)$ for any test input.

Platt scaling (Platt et al., 1999) is a parametric approach to calibration, unlike the other approaches. The non-probabilistic predictions of a classifier are used as features for a logistic regression model, which is trained on the validation set to return probabilities. More specifically, in the context of neural networks (Niculescu-Mizil & Caruana, 2005), Platt scaling learns scalar parameters $a, b \in \mathbb{R}$ and outputs $\hat{q}_i = \sigma(az_i + b)$ as the calibrated probability. Parameters a and b can be optimized using the NLL loss over the validation set. It is important to note that the neural network’s parameters are fixed during this stage.

4.2. Extension to Multiclass Models

For classification problems involving $K > 2$ classes, we return to the original problem formulation. The network outputs a class prediction $\hat{y}_i$ and confidence score $\hat{p}_i$ for each input $\mathbf{x}_i$. In this case, the network logits $\mathbf{z}_i$ are vectors, where $\hat{y}_i = \arg\max_k z_i^{(k)}$, and $\hat{p}_i$ is typically derived using the softmax function σ_{SM} :

$$\sigma_{\text{SM}}(\mathbf{z}_i)^{(k)} = \frac{\exp(z_i^{(k)})}{\sum_{j=1}^K \exp(z_i^{(j)})}, \quad \hat{p}_i = \max_k \sigma_{\text{SM}}(\mathbf{z}_i)^{(k)}.$$

The goal is to produce a calibrated confidence $\hat{q}_i$ and (possibly new) class prediction $\hat{y}'_i$ based on y_i , $\hat{y}_i$, $\hat{p}_i$, and $\mathbf{z}_i$.

² Because the validation dataset is finite, $\mathcal{S}$ is as well.